

MockStrive – An Intelligent Agentic AI Interview Trainer using IBM watsonx.ai

Problem Statement Document

1. Introduction

In today's competitive job market, technical knowledge alone is not enough to succeed in interviews. Students and job seekers must demonstrate problem-solving skills, communication, technical understanding, and confidence during interviews. However, many existing interview preparation platforms provide generic questions and limited feedback, making it difficult for candidates to prepare effectively.

With the advancement of Artificial Intelligence (AI), Large Language Models (LLMs), and Agentic AI, interview preparation can become more personalized, interactive, and practical. MockStrive is an AI-powered interview training platform built using IBM watsonx.ai, Meta Llama 3.3, and IBM Granite that helps candidates practice interviews, receive intelligent feedback, identify skill gaps, and continuously improve their interview performance.

2. Problem Statement

Preparing for technical interviews is challenging because every candidate has a different educational background, skill set, experience level, and career goal. Most existing interview preparation platforms rely on static question banks and predefined assessments that do not adapt to individual candidates or specific job roles.

As a result, candidates often spend time practicing irrelevant topics and receive only basic scores instead of meaningful feedback. They find it difficult to understand their strengths, identify weak areas, bridge skill gaps, and improve their confidence before attending real interviews.

There is a need for an intelligent interview training platform that analyzes a candidate's resume or profile, understands the target role and optional job description, generates personalized interview questions, evaluates every response, and provides actionable recommendations for improvement.

Such a system can make interview preparation more personalized, efficient, and accessible while helping candidates become better prepared for real-world recruitment processes.

3. Proposed Solution

MockStrive is an AI-powered Interview Trainer developed using React.js, Node.js, IBM watsonx.ai, Meta Llama 3.3, and IBM Granite.

The application collects candidate information including education, technical skills, projects, experience, target role, and an optional Job Description. Using this information, the system creates a personalized interview plan and conducts adaptive mock interviews.

The solution consists of multiple AI agents:

- **Profile Analysis Agent** analyzes the candidate's profile, identifies strengths, improvement areas, focus topics, and determines interview difficulty.
- **Interview Planner Agent** creates a personalized interview roadmap including interview mode, question distribution, and estimated duration.
- **Interview Question Agent** generates personalized questions for HR, Technical, DSA, Behavioral, AI/LLM, Mixed, and Company-Specific interview modes.
- **Evaluation Agent** evaluates every candidate response by analyzing technical correctness, communication quality, problem-solving ability, and overall performance.
- **Report Generation Agent** generates a comprehensive interview report including scores, strengths, weaknesses, learning topics, interview readiness, and personalized recommendations.

The analyzed profile is stored locally, allowing users to take multiple interviews without repeating profile analysis.

4. Objectives

The objectives of MockStrive are:

- Provide personalized interview preparation.
- Generate role-specific interview questions.

- Conduct multiple interview modes.
- Evaluate responses using AI.
- Improve interview confidence through repeated practice.
- Demonstrate Agentic AI using IBM watsonx.ai, Meta Llama 3.3, and IBM Granite.

5. Expected Outcomes

MockStrive helps candidates prepare for interviews through personalized practice and AI-generated feedback.

The platform enables users to:

- Practice different interview types.
- Identify strengths and improvement areas.
- Receive detailed performance reports.
- Improve confidence before real interviews.

The project also demonstrates the practical use of Agentic AI for education and career preparation.

6. Future Scope

The platform can be enhanced with ATS Resume Analysis, voice-based mock interviews, and company-specific interview preparation. It can also be integrated with college placement cells and training institutions to support a larger number of students.

7. Conclusion

MockStrive is an AI-powered interview training platform that uses IBM watsonx.ai with Meta Llama 3.3 and IBM Granite to provide personalized interview preparation, adaptive questioning, and AI-based performance evaluation.

The project demonstrates the practical use of Agentic AI to help candidates identify skill gaps, improve interview performance, and build confidence for real-world recruitment processes.